



ARTIFICIAL INTELLIGENCE

ment of outliers and missing data. Exploratory analysis is one such methodology used to study the differences of data in detail, thereby strengthening nutritional content.

Training a model. This step involves selecting the appropriate algorithm and representing data in the form of a model. The final purified data is split into two parts—training and test (proportion of data depends on prerequisite requirements). The first part (training data) is used to develop the model, whereas the second part (test data) is used as reference.

Evaluating the model. This step involves evaluation of the machine learning model you have chosen to implement. Second part of data (test data) is used to test the accuracy of the learning model. This step determines how precise the algorithm selected is, based on outcome.

There is also a better test to check accuracy of the model, which sees how the model performs on data that has not been used at all while building it.

Improving the performance. This step may involve choosing a different model altogether or even introducing more variables to improve the efficiency of the learning model. If the model is changed, then it again needs to be evaluated and its performance checked, which is why a lot of time needs to be spent in collecting and preparing data.

Tools for implementing machine learning

In order to implement machine learning on a system for any scenario, there are enough open source tools, software or frameworks available for you to choose from, based on your preference for a specific language or environment. Let us take a look at some of these.

Shogun. Shogun is one of the oldest and most venerable machine learning libraries available in the market. It was first developed in 1999 using C++, but now it is not limited to working in C++ only; rather, it can be used transparently in many languages and environments such as Java, C#, Python, Ruby, R, Octave, Lua and MATLAB. It is easy to use, and is quite fast at compilation and execution.

Weka. Weka was developed at University of Waikato in New Zealand. It collects a set of Java machine learning algorithms that are engineered specifically for data mining. This GNU GPLv3-licensed collection possesses a package system, which can be used to extend its functionality. It has both official and unofficial packages available.

Weka comes with a book that explains the software and the techniques used in it. While Weka is not aimed specifically at Hadoop users, it can be used with Hadoop as well, because of the set of wrappers that have been produced for the most recent versions of it. It does not support Spark, but Clojure users can also use Weka.

CUDA-Convnet. CUDA-Convnet is a machine learning library especially used for neural network applications. It is written in C++ in order to exploit Nvidia's CUDA GPU processing technology. It can even be used by those

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